Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 5

Hypothesis testing

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch05/code/ch05.R` (base R + MASS).

Sai Sridhar · October 20, 2026

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Answers at a glance

1	$b = 5.93 \times 10^{-4}$; BF $= 2.4,\ 3.8,\ 10.1,\ 35.2$ — never "decisive" (> 100), strong only for $N \geq 50$.
2	$\log_{10}$ BF ≈ 42 in favour of "vaccine arm has the lower rate"; insensitive to the prior.
3	BF(separate : common) $= 1.40$ ($c = 1$) but 0.14 ($c = 10$): the answer flips with the prior range.
4	(a) $\text{Prob}(\mu_1 < \mu_2) = 0.90$; (b) β_2 CI $(-1.8,\ 9.8)$; (c) $\text{Prob}(\mu_1 \neq \mu_2) = 0.53 \to 0.13$ as W grows.
5	$\tau = 1$: BF> 1 iff $\lvert y \rvert > 1.18$; $\tau = 10$: iff $\lvert y \rvert > 2.16$ (Lindley's paradox).
6	Mean opens: A 0.96, B 2.02, C 1.06; $\text{Prob}(\text{B best}) \approx 1$.
7	Prior $\text{Gamma}(0.44,\ 1359)$; BFDR 10% gives $t = 0.677$ and flags 8 regions.
8	Type I error 0.050–0.084 (> 0.05); power at $\lambda = 1.5$: 0.53 ($n = 10$) to 0.998 ($n = 100$); randomise to get exactly 0.05.
9	Most likely model drops `indus` and `age` (posterior probability 0.62).

A five-minute primer on the tools of this chapter

Background you need

1. Posterior probability of a hypothesis. For hypotheses about a parameter (e.g. $H_1 : \lambda \leq 1$ vs. $H_2 : \lambda > 1$), simply compute $\text{Prob}(H_k \mid \mathbf{Y})$ from the posterior.

2. Bayes factor. For two hypotheses (or models),

$$\text{BF}_{21} = \frac{\text{Prob}(H_2 \mid \mathbf{Y})/\text{Prob}(H_1 \mid \mathbf{Y})}{\text{Prob}(H_2)/\text{Prob}(H_1)} = \frac{\text{posterior odds}}{\text{prior odds}} = \frac{m_2(\mathbf{Y})}{m_1(\mathbf{Y})},$$

where $m_k(\mathbf{Y}) = \int f(\mathbf{Y} \mid \theta, H_k)\pi(\theta \mid H_k)\,\mathrm{d}\theta$ is the *marginal likelihood* (how well H_k, averaged over its prior, predicted the data). With equal prior probabilities, posterior odds = BF.

3. Interpreting a BF (Jeffreys' scale): 1–3 barely worth mentioning, 3–10 substantial, 10–30 strong, 30–100 very strong, > 100 decisive.

4. Sensitivity. For a *point* null ($\mu = 0$) the BF depends strongly on how spread out the alternative's prior is: a very vague alternative spreads its prior mass thinly and is penalised (**Lindley's paradox**). One-sided tests are much less sensitive.

5. Multiple testing: Bayesian false discovery rate. If we declare discoveries for all tests with $q_i = \text{Prob}(\text{effect}_i \mid \mathbf{Y}) > t$, the expected proportion of false discoveries among them is

$$\text{BFDR}(t) = \frac{\sum_i (1 - q_i)\mathbb{1}(q_i > t)}{\sum_i \mathbb{1}(q_i > t)}.$$

Choose the smallest t with $\text{BFDR}(t) \leq$ the target (e.g. 10%).

6. Variable selection (SSVS). Give each coefficient an inclusion indicator $\gamma_j \in \{0, 1\}$; the posterior inclusion probability $\text{Prob}(\gamma_j = 1 \mid \mathbf{Y})$ measures the evidence that covariate j belongs in the model.

Exercise 5.1 — A one-sided test for a Poisson rate

Exercise 1 (exact text from the book)

Assume that $Y|\lambda \sim \text{Poisson}(N\lambda)$ and $\lambda \sim \text{Gamma}(0.1, b)$. The null hypothesis is that $\lambda \leq 1$, and the alternative is that $\lambda > 1$. Select b so that the prior probability of the null hypothesis is 0.5. Using this prior, compute the posterior probability of the alternative hypothesis and the Bayes factor for the alternative relative to the null hypothesis for (a) $N = 10$ and $Y = 12$; (b) $N = 20$ and $Y = 24$; (c) $N = 50$ and $Y = 60$; (d) $N = 100$ and $Y = 120$. For which N and Y is there definitive evidence in favor of the alternative?

In plain words

The observed rate is always $Y/N = 1.2$; only the amount of data grows. Choose a prior with median 1, then see how the evidence for $\lambda > 1$ grows with N.

Background you need

- Poisson–gamma: $\lambda \mid Y \sim \text{Gamma}(0.1 + Y,\ b + N)$ (Chapter 2 primer).
- **Scaling**: if $\lambda \sim \text{Gamma}(a, b)$ then $b\lambda \sim \text{Gamma}(a, 1)$, so every quantile of λ is the corresponding $\text{Gamma}(a, 1)$ quantile divided by b.

Key idea

Prior probability 0.5 for $\lambda \leq 1$ means the prior *median* is 1. With equal prior odds, BF = posterior odds.

Solution

Step 1 — choose b.

The median of $\text{Gamma}(0.1, 1)$ is $m = \texttt{qgamma(0.5, 0.1, 1)} = 5.934 \times 10^{-4}$. The median of λ is m/b; setting it to 1 gives $b = m = 5.93 \times 10^{-4}$.

Why? With shape 0.1 the Gamma is extremely skewed: most of its mass is near 0 but it has a very long tail, so its median is tiny on the $\text{Gamma}(0.1, 1)$ scale.

Step 2 — posteriors.

$\lambda \mid Y \sim \text{Gamma}(0.1 + Y,\ 0.000593 + N)$, so $\text{Prob}(H_2 \mid Y) = \text{Prob}(\lambda > 1 \mid Y) = \texttt{pgamma(1, 0.1 + Y, b + N, lower.tail = FALSE)}$.

Step 3 — Bayes factors.

Prior odds $= 0.5/0.5 = 1$, so $\text{BF}_{21} = \frac{\text{Prob}(H_2|Y)}{1-\text{Prob}(H_2|Y)}$.

	N	Y	$\text{Prob}(\lambda > 1 \mid Y)$	BF_{21} (Jeffreys' scale)
(a)	10	12	0.707	2.4 (barely worth mentioning)
(b)	20	24	0.794	3.8 (substantial)
(c)	50	60	0.910	10.1 (strong)
(d)	100	120	0.972	35.2 (very strong)

Sanity check

With $N = 100$ the posterior is nearly $\text{N}(1.2,\ 1.2/100)$, so $\text{Prob}(\lambda > 1) \approx \Phi\left(\frac{0.2}{0.11}\right) = \Phi(1.83) = 0.966$, close to 0.972. ✓

Answer

$b = 5.93 \times 10^{-4}$. BF $= 2.4,\ 3.8,\ 10.1,\ 35.2$ for (a)–(d). The evidence grows with N, but none reaches "decisive" (BF > 100); (c) is strong and (d) very strong. If "definitive" means BF > 100, more data would be needed (about $N \approx 150$ at the same observed rate).

Takeaway & common mistakes

The same observed rate gives very different evidence depending on the sample size — always report BFs or posterior probabilities, not just point estimates.

Exercise 5.2 — Moderna: which arm has the lower infection rate?

Exercise 2 (exact text from the book)

Data from the initial Moderna vaccine trial in [9] are given in the table below.

Group	Infected (Y)	Total participants (n)
Placebo	185	14073
Vaccine	11	14134

Using a Bayes factor, test the hypotheses that H_1: the infection rate is lower in the placebo group versus H_2: the infection rate is lower in the vaccine group. Describe and justify your model and prior and test for sensitivity to the prior choice.

In plain words

Compare the hypotheses "placebo rate < vaccine rate" and "vaccine rate < placebo rate" with a Bayes factor, under several priors.

Background you need

- Binomial–beta: $\theta_g \mid Y_g \sim \text{Beta}(a + Y_g,\ b + n_g - Y_g)$, independently for the two arms.
- $\text{Prob}(\theta_V > \theta_P \mid \mathbf{Y}) = \int_0^1 f_P(x)\,\text{Prob}(\theta_V > x)\,\mathrm{d}x$, where f_P is the placebo posterior density.

Key idea

Use the *same* prior for both arms: then by symmetry $\text{Prob}(H_1) = \text{Prob}(H_2) = 0.5$ and $\text{BF}_{21} =$ posterior odds. The probability of H_1 is so small that Monte Carlo cannot see it; we integrate numerically on the log scale.

Solution

Model.

$Y_P \sim \text{Binomial}(14073, \theta_P)$, $Y_V \sim \text{Binomial}(14134, \theta_V)$ independently (randomised trial), $\theta_P, \theta_V \overset{\text{iid}}{\sim} \text{Beta}(a, b)$. $H_1 : \theta_P < \theta_V$; $H_2 : \theta_V < \theta_P$.

Computation.

$\text{Prob}(H_1 \mid \mathbf{Y}) = \int f_P(x)\text{Prob}(\theta_V > x)\,\mathrm{d}x$, evaluated on a 200,001-point grid with log-densities and a log-sum-exp (to avoid underflow); then $\log_{10}\text{BF}_{21} = \log_{10}\frac{1-\text{Prob}(H_1|\mathbf{Y})}{\text{Prob}(H_1|\mathbf{Y})}$.

Prior for both rates	$\text{Prob}(H_1 \mid \mathbf{Y})$	$\log_{10}\text{BF}_{21}$
$\text{Beta}(1,1)$ (uniform)	6.9×10^{-43}	42.2
$\text{Beta}(0.5,0.5)$ (Jeffreys)	3.2×10^{-43}	42.5
$\text{Beta}(1,99)$ (mean 1%, weak)	6.9×10^{-43}	42.2
$\text{Beta}(10,990)$ (mean 1%, strong)	6.5×10^{-38}	37.2

Why? Justification of the priors: the uniform and Jeffreys priors are objective defaults; the $\text{Beta}(1,99)$ and $\text{Beta}(10,990)$ priors encode the plausible belief that infection rates are around 1%, the latter as if 1,000 earlier participants had been observed in each arm — it deliberately pulls the two rates toward each other.

Sanity check

Independent check: for small rates $\text{Beta}(a, n) \approx \text{Gamma}(a, 1)/n$, giving $\text{Prob}(\theta_V > \theta_P) \approx \text{Prob}\big(\text{Beta}(12, 186) > \frac{14134}{28207}\big) = 1.2 \times 10^{-42}$ — the same order of magnitude as the exact 6.9×10^{-43}. ✓

Answer

Under every prior, $\text{BF}_{21} \approx 10^{37}$–$10^{42}$: overwhelming ("decisive" many times over) evidence that the infection rate is lower in the vaccine group. The conclusion is completely insensitive to the prior; even a strong prior pulling the rates together only changes the BF from 10^{42} to 10^{37}.

Takeaway & common mistakes

With 185 vs. 11 infections in arms of equal size, the data overwhelm any reasonable prior. Prior sensitivity matters when the evidence is modest, not here.

Exercise 5.3 — Mr. October: one rate or two?

Exercise 3 (exact text from the book)

Use the "Mr. October" data (Section 2.5) $Y_1 = 563$, $N_1 = 2820$, $Y_2 = 10$ and $N_2 = 27$. Compare the two models using Bayes factors:

$$H_1 : Y_1|\lambda_1 \sim \text{Poisson}(N_1\lambda_1) \text{ and } Y_2|\lambda_2 \sim \text{Poisson}(N_2\lambda_2)$$
$$H_2 : Y_1|\lambda_0 \sim \text{Poisson}(N_1\lambda_0) \text{ and } Y_2|\lambda_0 \sim \text{Poisson}(N_2\lambda_0).$$

Assume the Uniform(0,c) prior for all λ_j and compare the results for $c = 1$ and $c = 10$.

In plain words

Does Reggie Jackson have a separate World Series home-run rate (H_1), or one rate for all games (H_2)? Compute the Bayes factor for two prior ranges.

Background you need

$$\int_0^c \frac{e^{-N\lambda}(N\lambda)^Y}{Y!}\,\mathrm{d}\lambda = \frac{1}{N}\int_0^c \frac{N^{Y+1}\lambda^Y e^{-N\lambda}}{\Gamma(Y+1)}\,\mathrm{d}\lambda = \frac{1}{N}\,\text{pgamma}(c,\ Y+1,\ N),$$

because the integrand is a $\text{Gamma}(Y+1, N)$ density times $\frac{1}{N}$.

Key idea

Both marginal likelihoods have closed forms. H_1 has one more parameter, and each parameter's prior density is $1/c$, so BF_{12} scales roughly like $1/c$.

Solution

Step 1 — marginal likelihood of H_1 (separate rates).

The two integrals factorise:

$$m_1 = \prod_{j=1}^{2} \frac{1}{c}\int_0^c \text{Pois}(Y_j; N_j\lambda)\,\mathrm{d}\lambda = \prod_{j=1}^{2} \frac{\text{pgamma}(c, Y_j+1, N_j)}{c\,N_j}.$$

Step 2 — marginal likelihood of H_2 (common rate).

With $S = Y_1 + Y_2$ and $M = N_1 + N_2$, the product of the two Poisson PMFs is $\frac{N_1^{Y_1}N_2^{Y_2}}{Y_1!Y_2!}\lambda^S e^{-M\lambda}$, so

$$m_2 = \frac{1}{c}\cdot\frac{N_1^{Y_1}N_2^{Y_2}}{Y_1!\,Y_2!}\cdot\frac{\Gamma(S+1)}{M^{S+1}}\cdot\text{pgamma}(c, S+1, M).$$

Step 3 — results

(computed on the log scale):

c	BF_{12} (separate : common)	$\text{Prob}(H_1 \mid \mathbf{Y})$ (equal prior odds)
1	1.40	0.58
10	0.14	0.12

Why? All the pgamma terms are ≈ 1 (the likelihood lives far below c), so $\text{BF}_{12} \approx \frac{1}{c}\times$(a constant). Increasing c tenfold divides the BF by 10: the extra parameter of H_1 is "charged" for spreading its prior over a range ten times wider.

Answer

$c = 1$: $\text{BF}_{12} = 1.40$ (very weak support for separate rates). $c = 10$: $\text{BF}_{12} = 0.14$ (moderate support for a *common* rate). The conclusion reverses with the prior range — the Bayes factor is highly sensitive to a vague prior on the extra parameter.

Takeaway & common mistakes

Compare with Exercise 2.2, where the posterior probability that $\lambda_{WS} > \lambda_R$ was 0.95. Estimation ("which rate is bigger?") is robust to vague priors; model comparison with a vague prior on an extra parameter is not.

Exercise 5.4 — Comparing two deforestation strategies three ways

Exercise 4 (exact text from the book)

Researchers in Brazil are studying the effect of two intervention strategies to reduce deforestation. Their observational data consists of $n_1 = 7$ regions under strategy 1 and $n_2 = 9$ regions under strategy 2. Each observation was matched with a similar control region and the response is the difference in deforestation percent between the treated and control regions. The 16 differences are given in the table below.

Strategy	Differences (%)								
1	−11.5	−8.4	−6.8	−5.6	−0.4	1.9	2.8		
2	−7.0	−4.3	−3.0	−3.0	−0.1	0.0	2.2	4.3	10.8

(a) Let Y_{ij} be the ith observation under strategy j. The model is $Y_{ij}|\mu_j, \sigma_j \overset{indep}{\sim} \text{Normal}(\mu_j, \sigma_j^2)$. Select and justify uninformative priors, compute the posterior probabilities of the hypotheses $H_1 : \mu_1 < \mu_2$ versus $H_2 : \mu_2 < \mu_1$, and summarize the results of the test.

(b) Compare strategies using a simple linear regression for all the $i = 1, \ldots, n = 16$ observations. The likelihood is $Y_i|\boldsymbol{\beta}, \sigma^2 \overset{indep}{\sim} \text{Normal}(\beta_1 + \beta_2 X_i, \sigma^2)$ where $X_i = 0$ if the observation came from strategy 1 and $X_i = 1$ if it came from strategy 2. Select and justify uninformative priors, compute the 95% credible interval for β_2, and use this to compare strategies.

(c) Assume the regression model in (b) except set $\beta_2 = \gamma\delta$ with priors $\gamma \sim \text{Bernoulli}(0.5)$ and $\delta \sim \text{Normal}(0, W\sigma^2)$. Select and justify uninformative priors (try 2-3 values of W), compute the posterior probabilities of the hypotheses $H_1 : \mu_1 = \mu_2$ versus $H_2 : \mu_1 \neq \mu_2$, and summarize the results of the test.

(d) Comment on the pros and cons of these three tests.

In plain words

Negative differences mean less deforestation than the matched control. Compare the two strategies with (a) a one-sided test with separate variances, (b) a regression interval, and (c) a point-null test, and compare the approaches.

Background you need

- Jeffreys prior $\pi(\mu, \sigma^2) \propto 1/\sigma^2$ for a normal sample gives $\mu \mid \mathbf{Y} \sim t_{n-1}(\bar{Y},\ s^2/n)$.
- The same prior for regression gives $\beta_j \mid \mathbf{Y} \sim t_{n-p}(\hat{\beta}_j,\ \text{se}_j^2)$ — the least-squares estimate and its standard error.
- Point-null BF with $\delta \sim \text{N}(0, W\sigma^2)$, flat prior on β_1, $\pi(\sigma^2) \propto 1/\sigma^2$: with centred data $\tilde{y}, \tilde{x}$, $\text{BF}_{21} = (1 + W\tilde{x}^\top\tilde{x})^{-1/2}\left(\frac{S_2}{S_1}\right)^{-(n-1)/2}$, where $S_1 = \tilde{y}^\top\tilde{y}$ and $S_2 = S_1 - \frac{(\tilde{x}^\top\tilde{y})^2}{\tilde{x}^\top\tilde{x}+1/W}$.

Key idea

(a) and (b) are about *direction* and *size*; (c) asks whether the difference is *exactly zero*, which needs a proper prior on the difference — and its answer depends on W.

Solution

› Summary statistics.

$\bar{Y}_1 = -4.00$, $s_1 = 5.48$; $\bar{Y}_2 = -0.01$, $s_2 = 5.31$.

› (a) Separate Jeffreys priors.

$\pi(\mu_j, \sigma_j^2) \propto 1/\sigma_j^2$ for each strategy: objective, and lets the variances differ. Then $\mu_1 \mid \mathbf{Y} \sim t_6(-4.00,\ 5.48^2/7)$ and $\mu_2 \mid \mathbf{Y} \sim t_8(-0.01,\ 5.31^2/9)$. With 10^6 Monte Carlo draws:

$$\text{Prob}(H_1 : \mu_1 < \mu_2 \mid \mathbf{Y}) = 0.900, \qquad \mu_2 - \mu_1 : \text{ mean } 3.99,\ 95\%\text{ CI } (-2.48,\ 10.47).$$

Strategy 1 is probably more effective (it reduces deforestation by about 4 points more), with 90% posterior probability — suggestive, not conclusive.

› (b) Regression.

Prior $\pi(\beta_1, \beta_2, \sigma^2) \propto 1/\sigma^2$ (Jeffreys; reproduces least squares). $\hat{\beta}_2 = 3.99$, SE 2.71, so $\beta_2 \mid \mathbf{Y} \sim t_{14}(3.99,\ 2.71^2)$:

$$95\%\text{ credible interval for } \beta_2 :\ 3.99 \pm 2.145 \times 2.71 = (-1.83,\ 9.81), \qquad \text{Prob}(\beta_2 > 0 \mid \mathbf{Y}) = 0.918.$$

The interval contains 0, so at the 95% level we cannot say which strategy is better, though the evidence leans toward strategy 1.

› (c) Point-null test.

$\gamma = 0$ means $\mu_1 = \mu_2$. $\gamma \sim \text{Bernoulli}(0.5)$ gives equal prior probabilities; $\delta \sim \text{N}(0, W\sigma^2)$ scales the alternative to the noise level (W = prior variance of the difference in units of σ^2).

W	prior SD of δ	BF_{21}	$\text{Prob}(H_2 : \mu_1 \neq \mu_2 \mid \mathbf{Y})$
0.1	0.32σ	1.13	0.53
1	σ	1.05	0.51
10	3.2σ	0.45	0.31
100	10σ	0.15	0.13

With $W \leq 1$ the test is undecided; with vaguer priors it increasingly favours H_1 ($\mu_1 = \mu_2$) — Lindley's paradox again.

› (d) Pros and cons.

- **(a) One-sided posterior probability**: simple, interpretable ("90% chance strategy 1 is better"), robust to vague priors, allows unequal variances. But it does not allow "no difference" as a hypothesis.
- **(b) Credible interval**: shows both direction and plausible size of the effect. But it assumes equal variances, and "does the interval contain 0?" is a crude yes/no summary.
- **(c) Point-null Bayes factor**: gives real probability to "no difference", which may be scientifically meaningful. But the answer depends heavily on the arbitrary W, and an uninformative (huge) W always favours the null.

Answer

(a) $\text{Prob}(\mu_1 < \mu_2 \mid \mathbf{Y}) = 0.90$: strategy 1 probably reduces deforestation more. (b) 95% CI for β_2: $(-1.83,\ 9.81)$, includes 0 — no clear difference at 95%. (c) $\text{Prob}(\mu_1 \neq \mu_2 \mid \mathbf{Y})$ falls from 0.53 ($W = 0.1$) to 0.13 ($W = 100$): inconclusive and W-sensitive. (d) (a) and (b) are robust estimation-based summaries; (c) answers a sharper question but is prior-sensitive.

Takeaway & common mistakes

With 16 noisy observations no method gives a decisive answer — and the honest summary is "probably strategy 1, but more data are needed".

Exercise 5.5 — The Bayes factor for a normal mean

Exercise 5 (exact text from the book)

Equation 5.10 gives the Bayes factor for a test of a normal mean.

(a) For $\tau = 1$, plot the Bayes factor as a function of the observed value y and give a range of values of y that will lead to rejecting the null hypothesis.

(b) Repeat (a) with $\tau = 10$, and comment on the differences.

In plain words

One observation $Y \sim \mathrm{N}(\mu, 1)$; test $H_1 : \mu = 0$ vs. $H_2 : \mu \sim \mathrm{N}(0, \tau^2)$. See how the Bayes factor depends on y and on τ.

Background you need

Equation (5.10): $\mathrm{BF}(y) = (1+\tau^2)^{-1/2} \exp\left[\frac{y^2}{2}\left(\frac{\tau^2}{1+\tau^2}\right)\right]$ (BF of H_2 relative to H_1). It comes from $Y \sim \mathrm{N}(0,1)$ under H_1 and $Y \sim \mathrm{N}(0, 1+\tau^2)$ under H_2.

Key idea

"Reject H_1" when BF $> k$. Solve $\mathrm{BF}(y) > k$ for $|y|$:

$$|y| > \sqrt{\frac{1+\tau^2}{\tau^2} \cdot 2\Big(\log k + \tfrac{1}{2}\log(1+\tau^2)\Big)}.$$

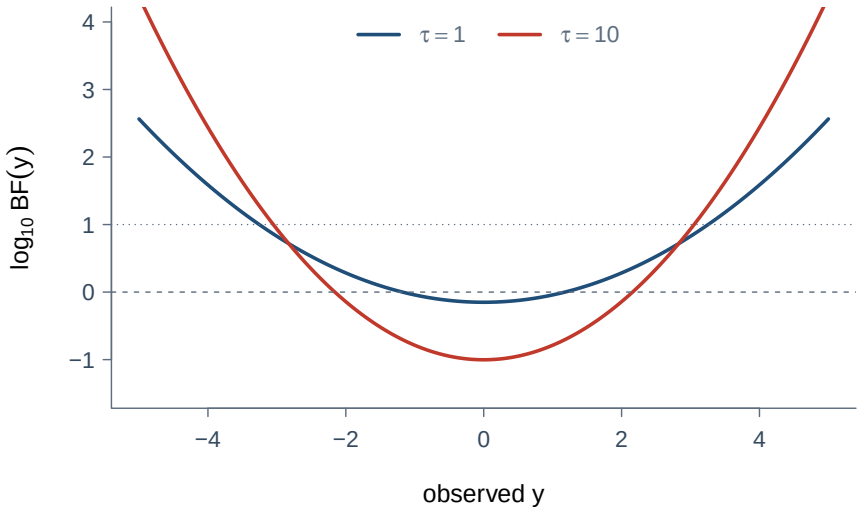
Solution

Derivation of the threshold.

Take logs of $\mathrm{BF}(y) > k$: $-\frac{1}{2}\log(1+\tau^2) + \frac{y^2}{2}\frac{\tau^2}{1+\tau^2} > \log k$. Move the first term over and multiply by $\frac{2(1+\tau^2)}{\tau^2}$ to get the boxed formula above.

Plot

($\log_{10}$ scale; dashed line BF $= 1$, dotted line BF $= 10$):



(a) $\tau = 1$.

$\mathrm{BF}(0) = 1/\sqrt{2} = 0.71$ (data at 0 mildly favour H_1).

- $\mathrm{BF} > 1$ (posterior odds favour H_2): $|y| > 1.18$.
- $\mathrm{BF} > 10$ (strong evidence; our rejection rule): $|y| > 3.26$.
- $\mathrm{BF} > 100$ (decisive): $|y| > 4.45$.

(b) $\tau = 10$.

$\mathrm{BF}(0) = 1/\sqrt{101} = 0.10$ (data at 0 *strongly* favour H_1).

- $\mathrm{BF} > 1$: $|y| > 2.16$. $\mathrm{BF} > 10$: $|y| > 3.05$. $\mathrm{BF} > 100$: $|y| > 3.74$.

Comments.

- A vaguer alternative ($\tau = 10$) needs a larger $|y|$ just to break even (2.16 vs. 1.18): it spreads its prior over a huge range and therefore predicts any particular moderate y poorly.
- But once $|y|$ is large, the BF grows faster for $\tau = 10$ (its curve is steeper), because $\frac{\tau^2}{1+\tau^2}$ is closer to 1.
- As $\tau \to \infty$, $\mathrm{BF}(y) \to 0$ for every fixed y — **Lindley's paradox**: an infinitely vague alternative can never win.

Answer

(a) $\tau = 1$: reject H_1 ($\mathrm{BF} > 10$) when $|y| > 3.26$ ($\mathrm{BF} > 1$ already for $|y| > 1.18$). (b) $\tau = 10$: reject when $|y| > 3.05$ ($\mathrm{BF} > 1$ only for $|y| > 2.16$). The vaguer prior penalises the alternative near $y = 0$ but rises faster for large $|y|$; in the limit $\tau \to \infty$ the null always wins.

Takeaway & common mistakes

For point-null tests, choose τ to reflect effect sizes you genuinely consider plausible — never "as large as possible".

Exercise 5.6 — A/B/C testing for a mobile health app

Exercise 6 (exact text from the book)

You are on the design team for a mobile health app. Your team is considering three new designs labelled A, B and C. To identify the optimal design, $n = 100$ users were randomly selected to be given each design and followed for a month. The table below gives the number of users in each treatment arm that opened the app y times for y between 0 and 6 (the maximum number of times the app was opened). For example, 100 users were given design A, and 40 of the 100 opened the app exactly once.

Design	0	1	2	3	4	5	6	Total
A	37	40	15	7	1	0	0	100
B	12	29	28	16	8	6	1	100
C	27	49	18	5	0	1	0	100

(a) Assuming the observations are Poisson counts, specify a likelihood and an objective Bayesian prior.

(b) Perform an A/B test. Plot the posterior distribution of the Poisson rate parameter (i.e., the mean response) for each group, and compute the posterior probability that each of the three designs is optimal.

In plain words

Which app design gets opened most often? Estimate each design's mean number of opens and the probability that each is the best.

Background you need

Poisson with Jeffreys prior (Exercise 2.19): $\pi(\lambda) \propto \lambda^{-1/2} \Rightarrow \lambda \mid \mathbf{Y} \sim \text{Gamma}(\sum Y_i + \frac{1}{2},\ n)$.

Key idea

Only the total number of opens per design matters. "Design g is optimal" = "λ_g is the largest"; its probability is the fraction of joint posterior draws in which λ_g wins.

Solution

(a) Likelihood and prior.

For user i with design g: $Y_{gi} \mid \lambda_g \overset{\text{iid}}{\sim} \text{Poisson}(\lambda_g)$; independent Jeffreys priors $\pi(\lambda_g) \propto \lambda_g^{-1/2}$ (objective and invariant to reparameterisation).

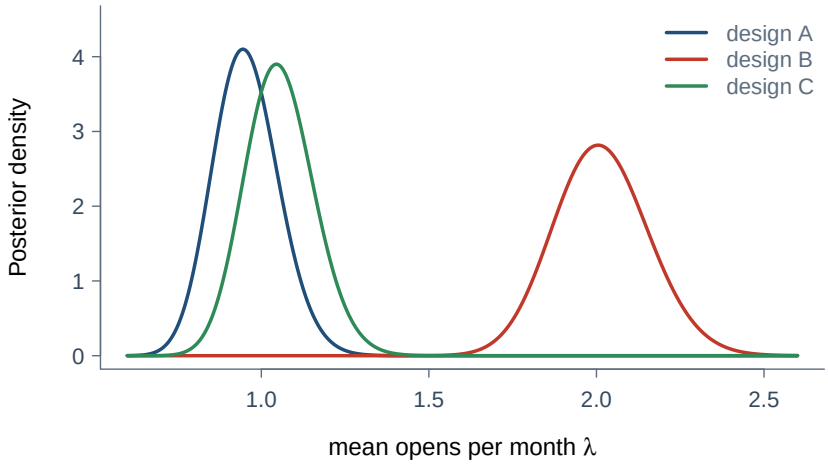
(b) Step 1 — totals.

A: $0\cdot37+1\cdot40+2\cdot15+3\cdot7+4\cdot1 = 95$; B: $29+56+48+32+30+6 = 201$; C: $49+36+15+0+5 = 105$.

(b) Step 2 — posteriors.

$\lambda_A \sim \text{Gamma}(95.5, 100)$, $\lambda_B \sim \text{Gamma}(201.5, 100)$, $\lambda_C \sim \text{Gamma}(105.5, 100)$.

Design	Posterior mean	95% interval
A	0.955	(0.773, 1.156)
B	2.015	(1.746, 2.303)
C	1.055	(0.864, 1.266)



› (b) Step 3 — probability each design is optimal

(10^6 joint draws): $\text{Prob}(\text{A best}) < 10^{-6}$, $\text{Prob}(\textbf{B best}) \approx 1$, $\text{Prob}(\text{C best}) < 10^{-6}$. (Between the other two, C beats A with probability ≈ 0.76.)

Sanity check

Is the Poisson assumption reasonable? Variance/mean ratios are A 0.94, B 0.98, C 0.78 — close to 1, so yes (C is slightly under-dispersed). ✓

Answer

(a) $Y_{gi} \overset{\text{iid}}{\sim} \text{Poisson}(\lambda_g)$ with Jeffreys priors $\pi(\lambda_g) \propto \lambda_g^{-1/2}$. (b) Posterior means: A 0.96, B 2.02, C 1.06 opens per month; design B is optimal with posterior probability ≈ 1.

Takeaway & common mistakes

"Probability of being the best" is the quantity that drives Bayesian bandit algorithms (Section 5.3): future users would be sent mostly to design B.

Exercise 5.7 — Thyroid cancer hot spots and the Bayesian false discovery rate

Exercise 7 (exact text from the book)

The national average rate of new thyroid cancer cases is around 13 per 100,000 people. An epidemiological study is monitoring certain regions for rates exceeding the national rate. The population size (N) and the number of new cases (Y) for each region for the most recent year are:

Region	N	Y	Region	N	Y
1	179611	20	11	30557	5
2	125453	18	12	81858	15
3	41646	7	13	103475	12
4	98826	22	14	166203	14
5	68642	10	15	52032	12
6	71844	6	16	85767	15
7	67702	9	17	145591	20
8	45954	9	18	139176	15
9	96325	18	19	87842	24
10	200201	27	20	97930	11

Separately for each region i, fit the model $Y_i|\lambda_i \sim \text{Poisson}(N_i\lambda_i)$ with prior $\lambda_i \sim \text{Gamma}(a, b)$, and compute the posterior probability that the region's rate λ_i exceeds the national average $\lambda_0 = 13/100000$, denoted $q_i = \text{Prob}(\lambda_i > \lambda_0|Y_i)$.

(a) Select a and b so that (approximately) the prior probability that $\lambda_1 > \lambda_0$ is 0.5, and the prior 0.95 quantile is $10\lambda_0$.

(b) Assuming that you will conclude the rate in region i exceeds the national average if $q_i > t$, select t to control the Bayesian false discovery rate at 10%. With this value, summarize the results.

In plain words

For each of 20 regions, compute the probability that its cancer rate exceeds the national rate; then choose a cut-off so that, on average, at most 10% of the regions we flag are false alarms.

Background you need

- Gamma scaling: quantiles of $\text{Gamma}(a, b)$ are $\text{Gamma}(a, 1)$ quantiles divided by b, so the *ratio* of two quantiles depends only on a.
- Bayesian FDR: primer, point 5.

Key idea

(a) Two conditions: median $= \lambda_0$ and 95th percentile $= 10\lambda_0$. The ratio $q_{0.95}/q_{0.5} = 10$ fixes a; then the median fixes b. (b) Rank the regions by q_i and flag from the top until the average of $1 - q_i$ among flagged regions would exceed 0.10.

Solution

(a) Step 1 — the shape a.

Solve $\frac{\texttt{qgamma(0.95, a, 1)}}{\texttt{qgamma(0.5, a, 1)}} = 10$ numerically: $a = 0.4395$.

(a) Step 2 — the rate b.

$b = \frac{\texttt{qgamma(0.5, a, 1)}}{\lambda_0} = 1359.4$. Check: $\text{Prob}(\lambda > \lambda_0) = 0.500$ and $q_{0.95} = 10.00\lambda_0$. ✓

Why? The prior is centred on the national rate but allows rates up to 10 times higher: a weakly informative, sceptical prior.

(a) Step 3 — q_i for each region.

$\lambda_i \mid Y_i \sim \text{Gamma}(0.4395 + Y_i,\ 1359.4 + N_i)$, $q_i = \text{Prob}(\lambda_i > \lambda_0 \mid Y_i)$. Sorted by q_i (rates per 100,000):

Region	Y/N	post. mean	q_i	flagged	Region	Y/N	post. mean	q_i	flagged
19	27.3	27.4	0.9993	✓	2	14.3	14.5	0.6531	
4	22.3	22.4	0.9890	✓	5	14.6	14.9	0.6290	
15	23.1	23.3	0.9622	✓	17	13.7	13.9	0.5902	
9	18.7	18.9	0.9234	✓	10	13.5	13.6	0.5698	
12	18.3	18.6	0.8911	✓	7	13.3	13.7	0.5174	
8	19.6	20.0	0.8678	✓	13	11.6	11.9	0.3369	
16	17.5	17.7	0.8578	✓	20	11.2	11.5	0.3034	
3	16.8	17.3	0.7319	✓	1	11.1	11.3	0.2345	
11	16.4	17.0	0.6765		18	10.8	11.0	0.2224	
					6, 14	8.4	8.6–8.8	≤ 0.12	

(b) Choosing t.

Flag top	6	7	**8**	9	10
threshold t	0.858	0.732	**0.677**	0.653	0.629
BFDR	0.061	0.073	**0.097**	0.122	0.145

Why? Example: flagging the top 8 regions, BFDR $= \frac{(1-0.9993)+(1-0.9890)+\cdots+(1-0.7319)}{8} = 0.097 \leq 0.10$; adding region 11 ($q = 0.677$) would raise it to 0.122.

So $t = 0.677$ (flag regions with $q_i > 0.677$).

Answer

(a) $\lambda \sim \text{Gamma}(0.44,\ 1359)$. (b) $t = 0.677$ controls the BFDR at 9.7%. Eight regions are flagged as exceeding the national rate: 19, 4, 15, 9, 12, 8, 16 and 3 (in decreasing order of evidence). Regions 19 and 4 (about 27 and 22 cases per 100,000) are nearly certain hot spots; for the others the evidence is moderate, and we expect less than one of the eight to be a false alarm.

Takeaway & common mistakes

Controlling the FDR accepts a few false alarms in exchange for power — appropriate for screening, where flagged regions are followed up anyway.

Exercise 5.8 — Frequentist properties of a Bayesian test

Exercise 8 (exact text from the book)

Assume the model $Y_i|\lambda \sim \text{Poisson}(\lambda)$ for $i = 1, \ldots, n$ and improper prior $\pi(\lambda) \propto 1$. The objective is to test the null hypothesis $H_1 : \lambda \leq 1$ versus the alternative hypothesis $H_2 : \lambda > 1$, and the test is to reject the null hypothesis if $\text{Prob}(\lambda > 1|Y_1, \ldots, Y_n) > 0.95$.

(a) Assuming n is fixed at 10, compute the probability of rejecting the null hypothesis for true λ ranging from 1 to 2.

(b) Assuming the true value of λ is 1, compute the Type I error of the test for n ranging from 10 to 100.

(c) Assuming the true value of λ is 1.5, compute the power of the test for n ranging from 10 to 100.

(d) How might you modify the test to guarantee that the Type I error is exactly 0.05?

In plain words

Study how often the Bayesian rule "reject if $\text{Prob}(\lambda > 1 \mid \text{data}) > 0.95$" rejects in repeated samples, for different true λ and sample sizes.

Background you need

- Flat prior + Poisson data: $\lambda \mid \mathbf{Y} \sim \text{Gamma}(S + 1,\ n)$ with $S = \sum Y_i$.
- $S \sim \text{Poisson}(n\lambda_{\text{true}})$ in repeated samples.

Key idea

The posterior depends on the data only through S, and $\text{Prob}(\lambda > 1 \mid S)$ increases with S. So the rule is "reject if $S \geq s_n$" for a critical value s_n, and $\text{Prob}(\text{reject}) = \text{Prob}(S \geq s_n)$ is an exact Poisson tail probability — no simulation needed.

Solution

› Step 1 — the critical value.

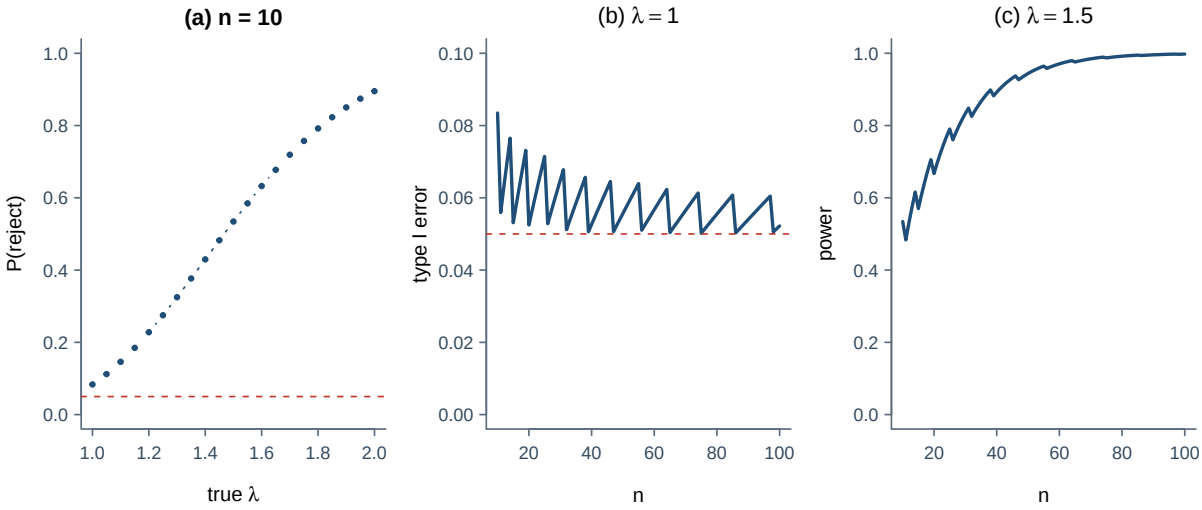
For each n, s_n is the smallest s with `pgamma(1, s + 1, n, lower.tail = FALSE)` > 0.95. For $n = 10$: $s_{10} = 15$.

› (a) $n = 10$.

$\text{Prob}(\text{reject} \mid \lambda) = \text{Prob}\big(\text{Poisson}(10\lambda) \geq 15\big)$:

true λ	1.0	1.2	1.4	1.6	1.8	2.0
Prob(reject)	0.084	0.228	0.430	0.633	0.792	0.895

› (b), (c) Type I error and power for $n = 10, \ldots, 100$.



Type I error ($\lambda = 1$) ranges from 0.050 to 0.084 (average 0.058) — *always at least* 0.05, with a saw-tooth pattern because S is discrete. Power at $\lambda = 1.5$: 0.53 ($n = 10$), 0.67 ($n = 20$), 0.94 ($n = 50$), 0.998 ($n = 100$).

Why? Why is the type I error too big? The flat prior $\pi(\lambda) \propto 1$ adds "one extra event" (posterior shape $S + 1$), nudging the posterior upward, so the rule rejects slightly too easily. The effect fades as n grows.

(d) Getting exactly 0.05.

- **Calibrate the cut-off**: for each n, choose s_n from the null distribution $S \sim \text{Poisson}(n)$ rather than from the 0.95 posterior rule; because S is discrete this gives type I error ≤ 0.05, and
- **randomise at the boundary**: reject if $S > s_n^*$, and reject with probability γ if $S = s_n^*$, with $\gamma = \frac{0.05 - \text{Prob}(S > s_n^*)}{\text{Prob}(S = s_n^*)}$. This makes the type I error exactly 0.05.

(Using the Jeffreys prior $\lambda^{-1/2}$ instead of the flat prior brings the error much closer to 0.05, but not exactly.)

Answer

(a) Rejection probability rises from 0.08 ($\lambda = 1$) to 0.90 ($\lambda = 2$). (b) Type I error 0.050–0.084, always above 0.05 (flat prior + discreteness). (c) Power at $\lambda = 1.5$ grows from 0.53 ($n = 10$) to 0.998 ($n = 100$). (d) Calibrate the cut-off to the null distribution of S and randomise at the boundary.

Takeaway & common mistakes

Bayesian procedures can (and often should) be checked for their frequentist behaviour — this is how sample sizes for Bayesian trials are planned (Section 5.4).

Exercise 5.9 — Stochastic search variable selection for Boston housing

Exercise 9 (exact text from the book)

Download the Boston Housing Data in R from the Boston dataset. The response is medv, the median value of owner-occupied homes, and the other 13 variables are covariates that describe the neighborhood. Use stochastic search variable selection (SSVS) to compute the most likely subset of the 13 covariates to include in the model and the marginal probability that each variable is included in the model. Clearly describe the model you fit including all prior distributions.

In plain words

Which of the 13 neighbourhood variables really belong in the regression for house values? Use SSVS to get each variable's inclusion probability and the best subset.

Background you need

Collapsed SSVS. With a conjugate prior the regression coefficients can be integrated out, leaving the marginal likelihood $m(\mathbf{Y} \mid \boldsymbol{\gamma})$ of each subset $\boldsymbol{\gamma}$ in closed form. The Gibbs sampler then updates each γ_j from

$$\text{Prob}(\gamma_j = 1 \mid \boldsymbol{\gamma}_{-j}, \mathbf{Y}) = \frac{m(\mathbf{Y} \mid \gamma_j = 1, \boldsymbol{\gamma}_{-j})}{m(\mathbf{Y} \mid \gamma_j = 1, \boldsymbol{\gamma}_{-j}) + m(\mathbf{Y} \mid \gamma_j = 0, \boldsymbol{\gamma}_{-j})} \quad (\text{prior } \gamma_j \sim \text{Bernoulli}(0.5)).$$

Key idea

With only $2^{13} = 8{,}192$ possible subsets we can also compute the *exact* posterior of every model and check the SSVS sampler against it.

Solution

Model and priors.

$$\text{medv}_i = \beta_0 + \sum_{j=1}^{13} \gamma_j \beta_j Z_{ij} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2),$$

with Z_{ij} the standardised covariates and

$$\pi(\beta_0) \propto 1, \qquad \beta_j \mid \sigma^2 \overset{\text{iid}}{\sim} \text{N}(0,\ W\sigma^2)\ (W = 1), \qquad \pi(\sigma^2) \propto 1/\sigma^2, \qquad \gamma_j \overset{\text{iid}}{\sim} \text{Bernoulli}(0.5).$$

Why? $W = 1$ says an included variable's effect (per SD) is typically of the order of the noise SD ($\approx \$4{,}700$) — realistic for house prices; $\gamma_j \sim \text{Bernoulli}(0.5)$ makes all 8,192 subsets equally likely a priori.

Marginal likelihood of a subset.

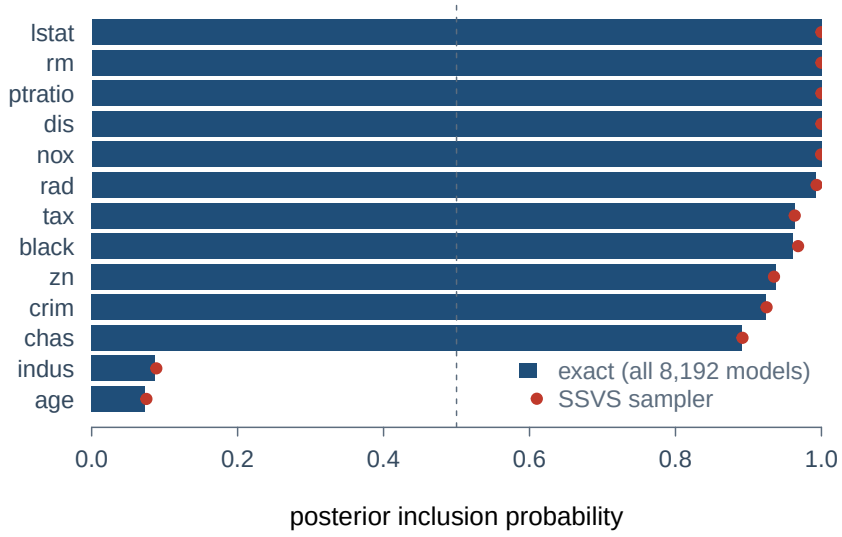
Integrating out β_0, $\boldsymbol{\beta}_\gamma$ and σ^2 (centred $\tilde{\mathbf{y}}$, $k = \sum \gamma_j$, $\mathbf{A} = \mathbf{Z}_\gamma^\top \mathbf{Z}_\gamma + \mathbf{I}/W$):

$$\log m(\mathbf{Y} \mid \boldsymbol{\gamma}) = -\tfrac{k}{2} \log W - \tfrac{1}{2} \log |\mathbf{A}| - \tfrac{n-1}{2} \log \left(\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}} - \tilde{\mathbf{y}}^\top \mathbf{Z}_\gamma \mathbf{A}^{-1} \mathbf{Z}_\gamma^\top \tilde{\mathbf{y}} \right) + \text{const.}$$

SSVS sampler.

6,000 sweeps (1,000 burn-in), each updating all 13 γ_j in random order.

Variable	SSVS	exact	Variable	SSVS	exact
crim	0.925	0.924	dis	1.000	1.000
zn	0.935	0.937	rad	0.993	0.992
indus	**0.089**	0.087	tax	0.963	0.964
chas	0.892	0.891	ptratio	1.000	1.000
nox	1.000	1.000	black	0.968	0.961
rm	1.000	1.000	lstat	1.000	1.000
age	**0.075**	0.073			



› Most likely subsets.

Posterior prob.	Model
0.615	all except `indus` and `age`
0.079	all except `indus`, `age` and `chas`
0.056	all except `age`
0.048	all except `indus`
0.045	all except `indus`, `age` and `crim`

Sanity check

The SSVS chain visits the top model 61.8% of the time, matching its exact posterior probability 0.615, and every inclusion probability agrees with the exact enumeration to within 0.007. ✓

Answer

Model: standardised linear regression with $\beta_j \mid \sigma^2 \sim \mathrm{N}(0, \sigma^2)$ for included variables, flat intercept, $\pi(\sigma^2) \propto 1/\sigma^2$, $\gamma_j \sim \text{Bernoulli}(0.5)$. The most likely subset (posterior probability 0.62) contains all variables except `indus` and `age`. Inclusion probabilities: `indus` 0.09, `age` 0.07, `chas` 0.89, `crim` 0.92, `zn` 0.94, `tax` 0.96, `black` 0.96, the rest ≥ 0.99.

Takeaway & common mistakes

This agrees with Exercise 4.2: `indus` and `age` were exactly the coefficients whose intervals contained 0 and which the LASSO prior shrank to zero.